

Project - High Level Design

On

Marketing Content Generator

Course Name: Generative AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. INTRODUCTION

The **Marketing Content Generator** is a Generative AI-based system designed to automate the creation of marketing content such as social media posts, ad copies, email campaigns, product descriptions, and blog ideas.

In traditional marketing workflows, content creation requires human creativity, time, and effort. This often leads to delays, inconsistency, and high dependency on skilled content writers.

The proposed system uses Generative AI to automatically generate high-quality, context-aware, and customizable marketing content based on user inputs such as product details, target audience, tone, and platform.

This system improves productivity, reduces manual effort, and ensures consistency in branding and messaging.

1.1. Scope of the document

This document presents the High Level Design (HLD) of the Marketing Content Generator system.

It includes:

- System architecture.
- Application design.
- Data flow.
- Components interaction.
- Interfaces.
- Non-functional requirements.

This document focuses only on conceptual design, not implementation details.

1.2. Intended Audience

This document is intended for:

- Students and project team members for understanding system design.
- Faculty and evaluators for assessing project structure.
- Developers and designers for future implementation.
- Reviewers for analyzing GenAI-based system architecture.

1.3. System overview

- The Marketing Content Generator is an AI-powered system that generates marketing content based on user input.
- **Key Functionality:**
- Accepts input. (product, audience, tone, platform)
- Uses GenAI model to generate content.
- Customizes output. (formal, casual, creative, etc.)
- Supports multiple formats. (ads, captions, emails, blogs)
- **Example:**
- User Input:
"Create Instagram caption for protein powder for gym audience"
- Output:
Engaging caption with hashtags and CTA.

2. System Design

The **Marketing Content Generator** is designed using a **modular, layered, and scalable architecture**. The main goal of this design is to ensure that each part of the system performs a specific responsibility without interfering with other components.

This design follows **Separation of Concerns**, meaning:

- Input handling.
- AI processing.
- Output formatting.

This makes the system:

- Easy to maintain.
- Easy to upgrade.
- Easy to debug.

2.1. Application Design

The application is divided into multiple modules, each responsible for a specific task in the content generation pipeline.

1. User Interface (UI Layer)

This is the **entry point of the system**.

Responsibilities:

- Accepts user input such as:
 - Product name
 - Target audience
 - Platform (Instagram, Email, Ads)
 - Tone (formal, casual, creative)
- Displays generated content

Example Input:

"Generate Instagram caption for gym audience for protein powder in a motivating tone"

Why important?

- Ensures smooth interaction between user and system
- Makes system user-friendly

2. Prompt Processor (Pre-processing Layer)

This module converts raw user input into a **structured AI-friendly prompt**.

Responsibilities:

- Cleans input text
- Adds missing context
- Converts simple input into detailed instruction

Example Transformation:

User Input:

"Instagram caption for shoes"

Processed Prompt:

"Generate a catchy Instagram caption for sports shoes targeting young audience with energetic tone including hashtags and call-to-action"

Why important?

- AI works better with structured prompts
- Improves output quality

3. Content Generation Engine (Core AI Layer)

This is the **heart of the system**.

Responsibilities:

- Uses Generative AI (LLM) to create content
- Understands context, tone, and intent
- Generates human-like marketing text

Capabilities:

- Ad copy generation
- Social media captions
- Email marketing content
- Blog ideas

Why important?

- Main intelligence of the system
- Directly affects quality of output

4. Tone & Style Controller (Customization Layer)

This module modifies generated content based on tone and branding.

Supported tones:

- Formal
- Casual
- Professional
- Emotional

Example:

Same product, different tone:

- Casual: "Grab yours now before it's gone!"
- Professional: "Enhance your performance with our premium product."

Why important?

- Personalization
- Brand consistency

5. Template Engine (Structuring Layer)

This module applies predefined structures depending on content type.

Examples:

- Email format → Subject + Body + CTA
- Instagram → Caption + Hashtags
- Ads → Headline + Description

Why important?

- Ensures professional formatting
- Saves time

6. Output Formatter (Presentation Layer)

Final step before showing output to user.

Responsibilities:

- Improves readability
- Removes unwanted text
- Formats spacing, bullet points, etc.

Why important?

- Clean and ready-to-use output
- Better user experience

2.2. Process Flow

The system follows a structured pipeline:

- User enters requirement.
- Input is received by UI.
- Prompt Processor refines input.
- Structured prompt sent to AI Engine.
- AI generates raw content.
- Tone Controller modifies style.
- Template Engine structures content.
- Formatter cleans output.
- Final content displayed.

2.3. Information Flow

This describes **how data moves inside the system.**

- User Input → UI.
- UI → Prompt Processor.
- Prompt → AI Engine.
- AI Output → Tone Controller.
- Modified Content → Template Engine.
- Structured Output → Formatter.
- Final Output → UI.
- Each module only communicates with the next module → ensures modularity.

2.4. Components Design

Each component is independent and loosely coupled.

User Interface

- Input collection.
- Output display.

Prompt Processor

- Converts input → structured prompt.
- Enhances clarity.

AI Content Generator

- Uses LLM.
- Generates content.

Tone Controller

- Adjusts emotional and stylistic tone.

Template Engine

- Applies structure based on platform.

Formatter

- Final cleanup and presentation.

Design Advantage:

If you want to upgrade AI model → only change AI Engine

If you want new format → update Template Engine.

2.5. Key Design Considerations

1. Modularity

- Each module works independently
→ Easy maintenance.

2. Scalability

System can handle:

- More users.
- More content types.

3. Flexibility

Supports:

- Different industries.
- Multiple platforms.

4. Performance

- Fast response using optimized prompts.
- Lightweight architecture.

5. Customization

- User controls tone, style, audience.

6. Reliability

- Consistent output.
- Minimal errors.

3. Data Design

Data Design defines how data is **structured, stored, accessed, and managed** within the system.

In the **Marketing Content Generator**, the system is designed to be **lightweight and efficient**, so it primarily uses **in-memory data handling** instead of a heavy database.

The focus is on:

- Fast processing.
- Temporary data usage.
- Privacy and simplicity.

3.1 Data Model

The system revolves around a few **core data entities**. These entities represent the information required to generate marketing content.

1. User Input Entity

This entity captures all inputs provided by the user.

Attributes:

- product_name → Name of the product/service
- target_audience → e.g., students, gym users, professionals
- platform → Instagram, Email, Ads, Blog
- tone → Formal, Casual, Emotional, Professional
- keywords → Optional keywords for SEO/content focus

2. Prompt Entity

This is the **processed version of user input** that is sent to the AI model.

Attributes:

- prompt_text → Structured instruction for AI
- context_info → Additional instructions (hashtags, CTA, etc.)

3. Generated Content Entity

This stores the output produced by the AI model.

Attributes:

- content_text → Final generated content
- content_type → Caption, Email, Ad Copy
- length → Short / Medium / Long

3.2 Data Access Mechanism

The system uses **in-memory data structures** instead of databases.

How Data is Stored

- Lists
- Dictionaries (Key-Value pairs)

Why In-Memory?

- Faster processing
- No database setup required
- Perfect for college-level project
- Reduces system complexity

Data Flow in Memory

1. Input stored temporarily
2. Converted into prompt
3. Passed to AI
4. Output stored temporarily
5. Displayed to user
6. Data cleared

3.3 Data Retention Policies

This defines **how long data is stored**.

Current System Policy

- Data is **temporary**
- Stored only during execution
- Deleted immediately after output generation

Why This Approach?

- Ensures **user privacy**
- Reduces **memory usage**
- Avoids **data security risks**

Future Enhancement

- Save user history
- Store previous content
- Build personalized suggestions

4. Interfaces

The interface defines how users interact with the Marketing Content Generator system. In the current design, a **Command Line Interface (CLI)** is used, where users provide inputs such as product details, target audience, platform, and tone, and receive generated marketing content as output. The interface is designed to be simple, intuitive, and easy to use, requiring minimal technical knowledge. It acts as a communication bridge between the user and internal system components like the prompt processor and AI engine. For future enhancements, the system can support a **web-based interface**, mobile application, or REST APIs to enable integration with external marketing tools and platforms.

5. State and Session Management

State and session management defines how the system handles user interactions and execution data. The Marketing Content Generator is designed as a **stateless system**, meaning each request is processed independently without storing previous interactions. When a user submits input, the system processes it, generates content, and returns the output without maintaining any session data. This approach simplifies system design, improves scalability, and allows multiple users to use the system simultaneously without interference. Since no user-specific data is stored, it also enhances privacy and security. In future versions, session management can be introduced to support features like user history, saved content, and personalized recommendations.

6. Caching

Caching refers to the temporary storage of frequently used or previously generated data to improve system performance and reduce response time. In the current design of the Marketing Content Generator, caching is not implemented because the system processes each request independently using real-time AI generation. However, caching can be introduced in future versions to store commonly used prompts, templates, or previously generated content. This would help in reducing repeated computations and API calls, thereby improving efficiency and lowering response time. For example, frequently requested marketing captions or templates can be quickly retrieved from cache instead of being regenerated, enhancing overall system performance and user experience.

7. Non-Functional Requirements

Non-functional requirements define the **quality attributes** of the Marketing Content Generator system. Unlike functional requirements, which describe what the system does, non-functional requirements describe **how well the system performs** under different conditions. These requirements ensure that the system is efficient, secure, reliable, and user-friendly.

7.1 Security Aspects

Security ensures that user input and generated content are handled safely.

In the current system:

- No sensitive user data is stored
- All inputs are processed temporarily in memory
- No long-term data retention reduces risk of data leaks

Basic security measures include:

- Input validation to prevent malicious prompts
- Safe API usage for AI model interaction
- Limiting unauthorized access to system components

Future Enhancements:

- User authentication and authorization
- Secure API keys and encryption
- Role-based access control

7.2 Performance Aspects

Performance defines how quickly and efficiently the system responds to user requests.

Current design ensures:

- Fast response time using lightweight architecture
- Real-time content generation with optimized prompts
- Minimal processing delay due to in-memory operations

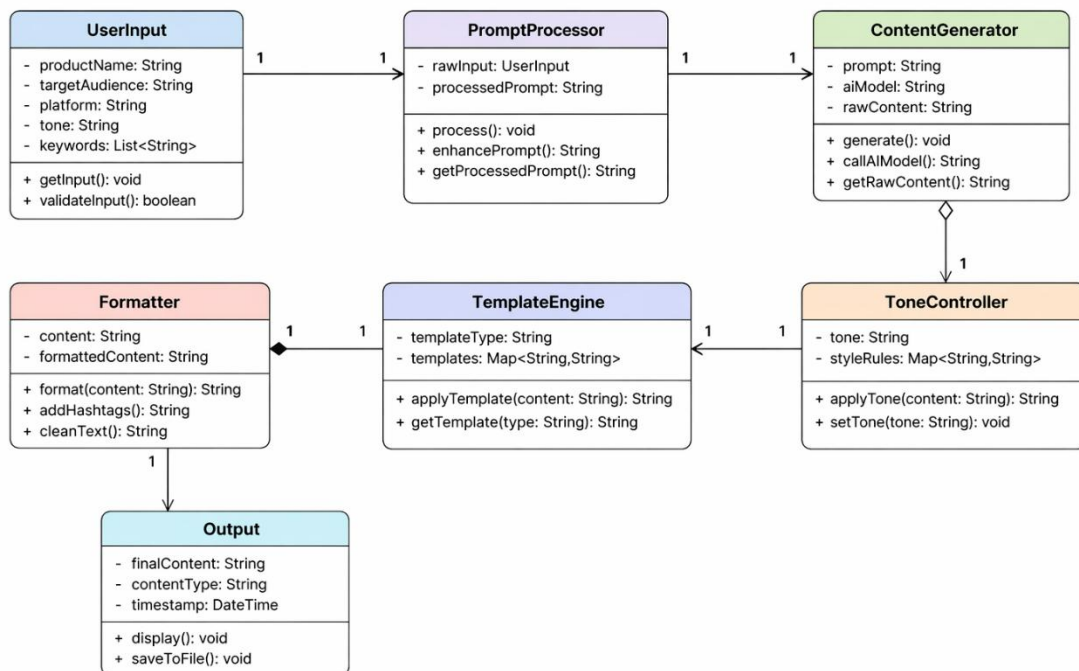
Factors affecting performance:

- AI model response time
- Complexity of user prompt
- Network latency (if using APIs)

8. Diagram & Flowcharts :

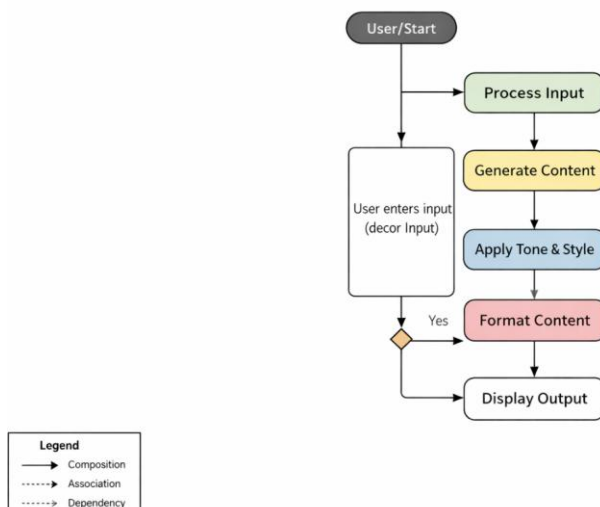
8.1 Class Diagram

Class Diagram



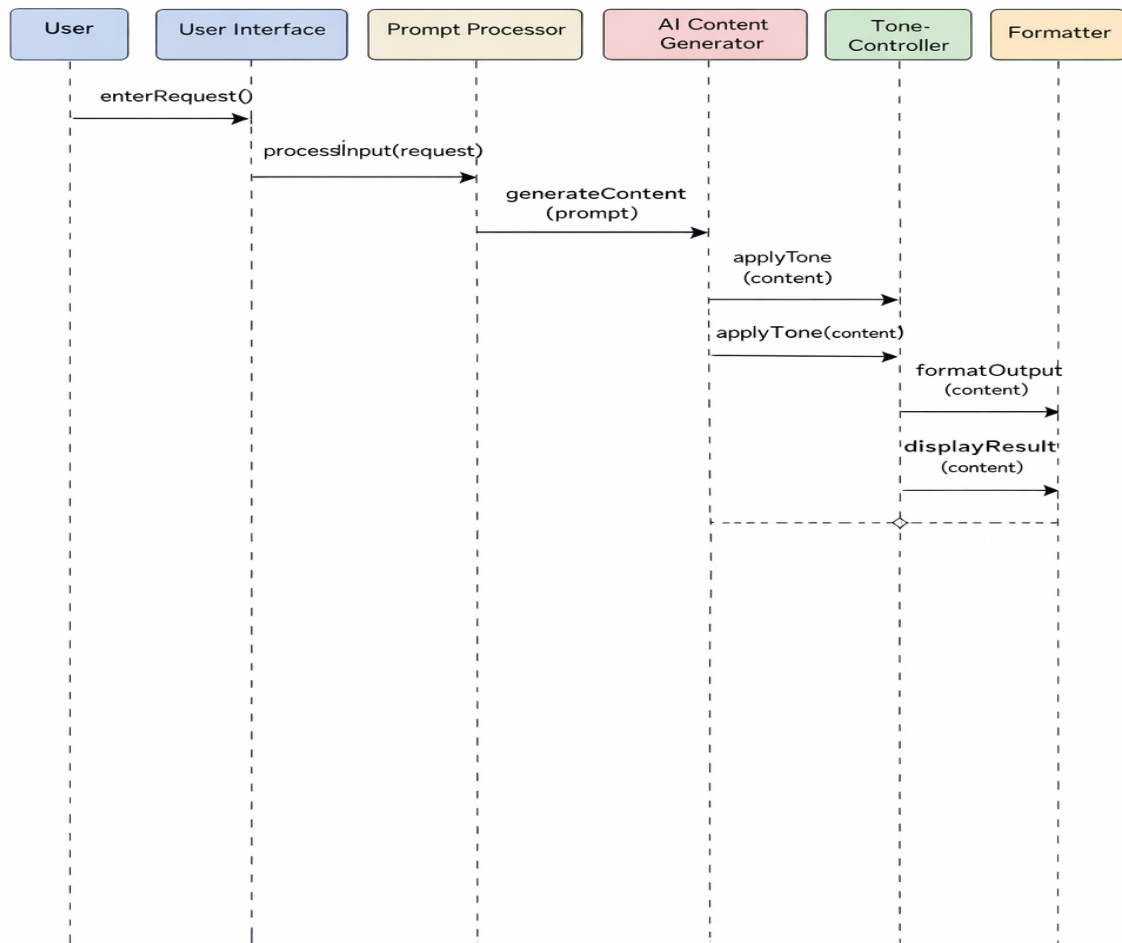
8.2 Activity Diagram

Activity Diagram – Marketing Content Generator

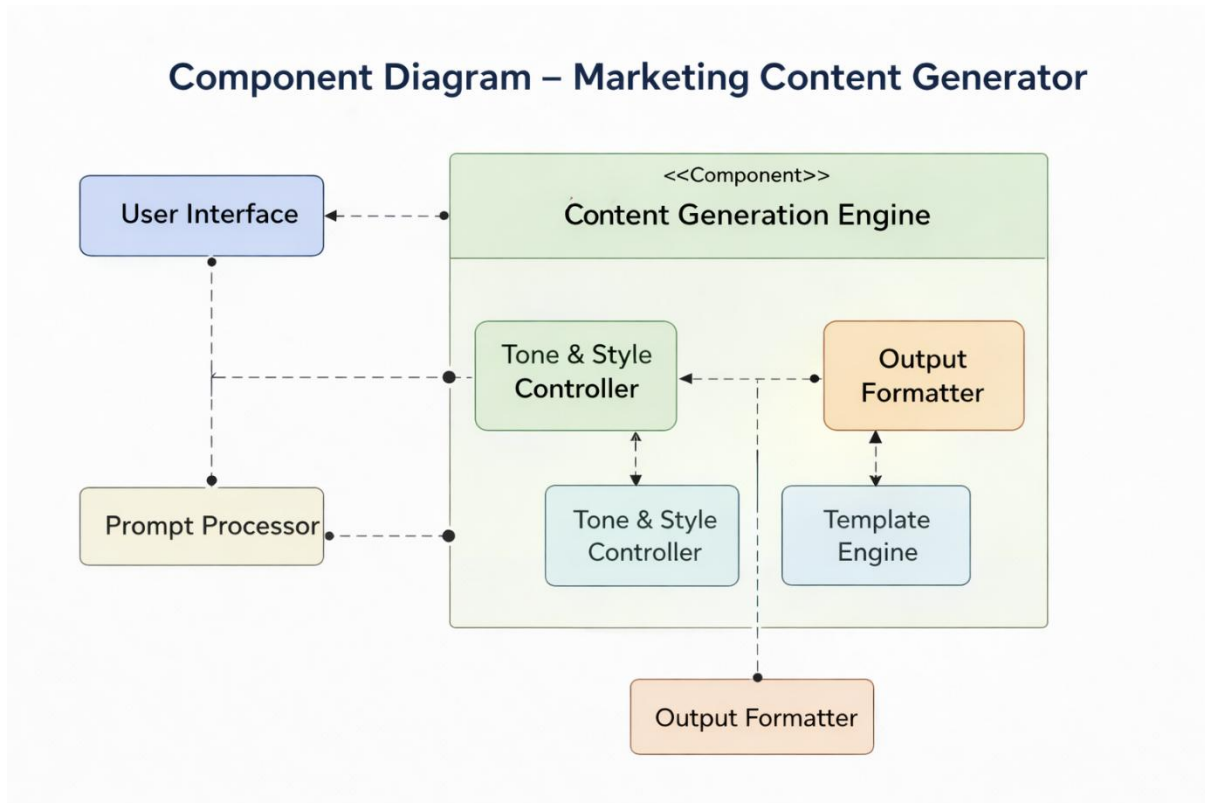


8.3 Sequence Diagram

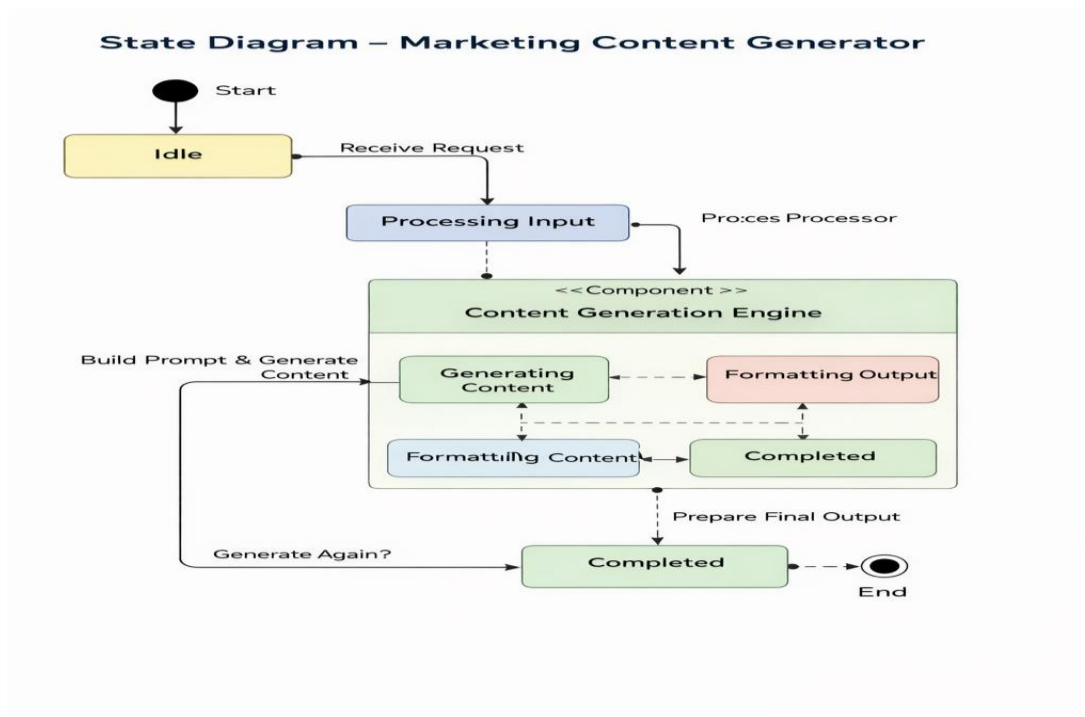
Sequence Diagram – Marketing Content Generator



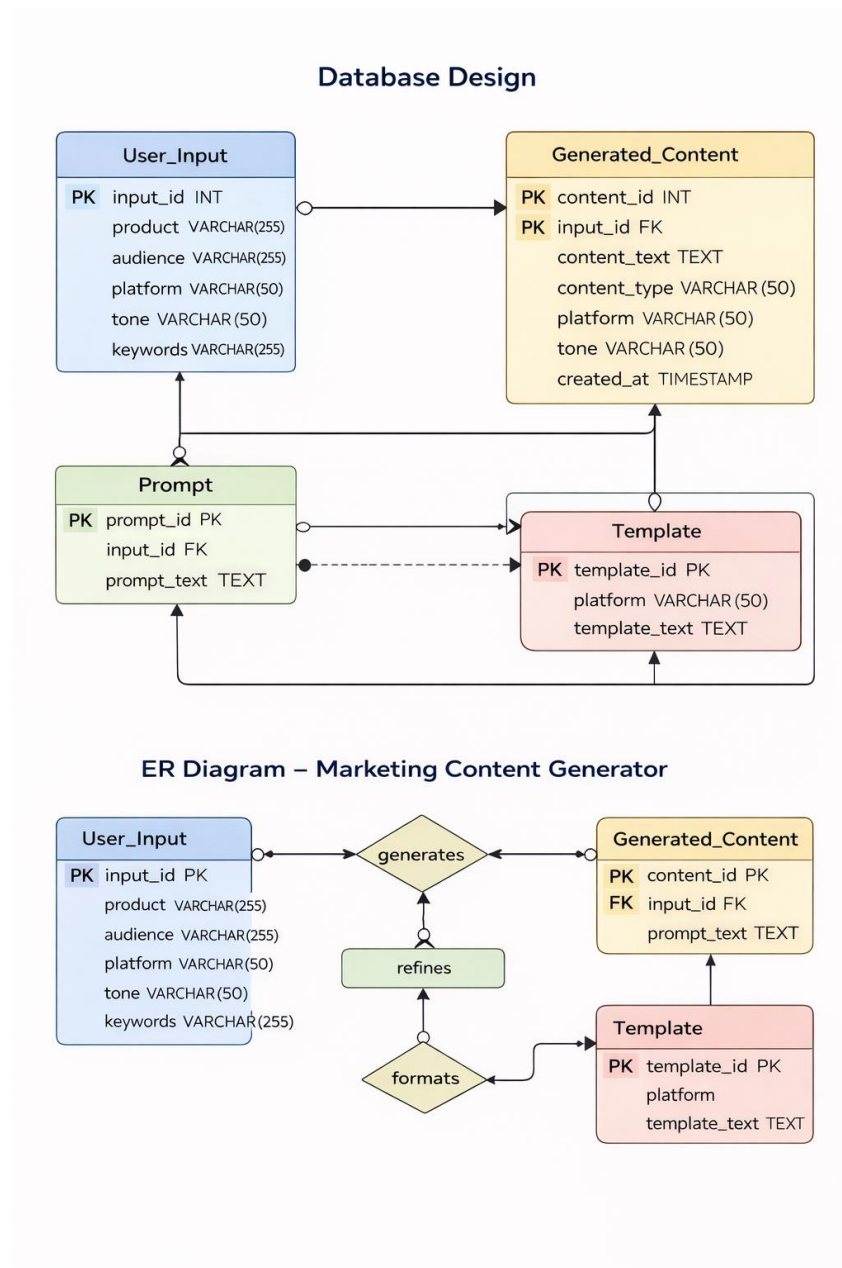
8.4 Component Diagram



8.5 State Diagram



8.6 Database Design & ER Diagram



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